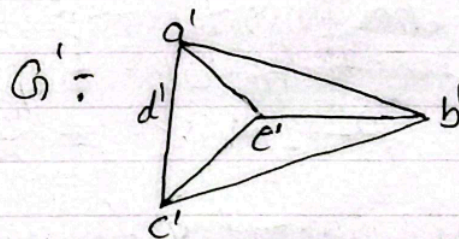
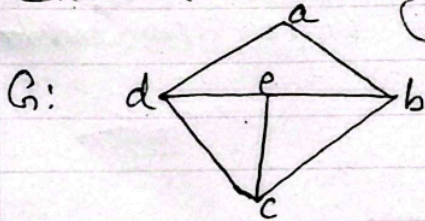


BSc. CS27
2nd sem

Assignment Discrete Structure.

Submission Date
up to 2023/05/26

- ① What is Logical equivalence? State and prove De Morgan's Laws using laws of logic.
- ② Explain the Euler path and Euler Circuit with the help of a diagram. State the necessary and sufficient conditions for Euler Circuits and paths.
- ③ Derive the recurrence relation for the Tower of Hanoi and solve it using an iterative approach.
- ④ Find all solutions of the recurrence relation
$$a_n = 5a_{n-1} - 6a_{n-2} + 42 \cdot 4^n$$
With Initial Conditions $a_1 = 56$ & $a_2 = 278$
- ⑤ What are the applicable fields of graph theory? Explain Hamilton graph with properties.
- ⑥ Show that the graphs G & G' are isomorphic.



- ⑦ Short notes on:
 - a) Multigraph
 - b) Pseudograph
 - c) Complete graph

①

- ⑧ What do you mean by minimum spanning tree?
Find the minimum spanning tree of graph (any)
using
- Kruskal algorithm
 - Prim's algorithm.
- ⑨ prove that following propositions are tautology.
 (a) $\neg (p \wedge q) \vee q$ (b) $p \rightarrow (p \vee q)$
- ⑩ Show that the 't' is valid conclusion from the premises
 $p \rightarrow q, q \rightarrow r, r \rightarrow s, \neg s$ and $p \vee t$.
- ⑪ Define regular & isomorphic graph. Differentiate between Walk, path and Trail in graph.
- ⑫ Prove the validity of the following argument "If I get the job and work hard, then I will get promoted. If I get promoted, then I will be happy. I will not be happy". Therefore "either I will not get job or I will not work hard".
- ⑬ Explain the importance of predicate logic in Computer Science. Use truth table to prove.
 $(p \rightarrow q) \vee \neg p \equiv \neg p \vee q$
- ⑭ State the rules of Inference for propositional logic.
- ⑮ Differentiate between direct proof and indirect proof with example.

(16) Find the solution to recurrence relation.
 $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ With initial
Condition $a_0 = 2, a_1 = 5$ & $a_2 = 15$.

(17) Derive the explicit formula for Fibonacci Series.

(18) Use mathematical induction to show $7^{n+2} + 8^{2n+1}$
is divisible by 57 for $n \geq 0$.

(19) (a) Use direct proof to prove "If x is odd then x^{2^k} is
also odd."

(b) Show that if $n^3 + 2n + 1$ is odd then n is even
by contrapositive.

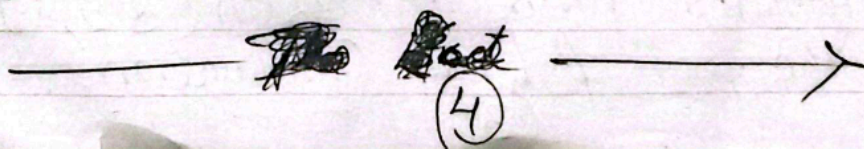
(20) Construct an argument using rule of inference to show
that the hypothesis "If it does not rain or if it is
not foggy then sailing race will be held and the life
saving demonstration will go on," "If the sailing race
is held then the trophy will be awarded," and "The trophy
was not awarded" imply the conclusion "It rained".
You are required to show each step and give reasons
for those steps.

(21) Using mathematical induction prove that
 $1^2 + 2^2 + 3^2 + \dots + n^2 = \{n(n+1)(2n+1)\}/6$

(22) How do you define Euclidean Algorithm and Extended
Euclidean Algorithm. Find the GCD (83, 19) and
Express in the form of linear combination.

(3)

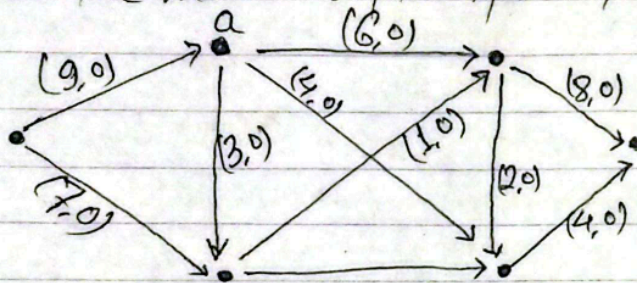
- 23) Define closure of a relation. Let $A = \{1, 2, 3, 4\}$ and Let $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$. Find the transitive closure of R using Warshall's Algorithm.
- 24) Use Mathematical Induction to show that $8^n - 3^n$ is divisible by 5.
- 25) Define injective, bijective and surjective functions. Show that the function $f: R \rightarrow R$, given by function $f(x) = 2x + 7$, is one-one and onto.
- 26) Explain different types of quantifiers used in predicate logic.
- 27) Differentiate between Permutation & Combination. The 5 letters in the list 'L', 'M', 'N', 'O' and 'P' are to be arranged so that 'L' is the third letter in the list and 'M' is not next to 'L'. How many arrangements are there?
- 28) What is relation? Explain transitive, reflexive and equivalence relation with example.
- 29) Let $A = Z$, the set of integers and Let $R = \{(a, b) : a \leq b\}$. Is R reflexive, symmetric, anti-symmetric or transitive.
- 30) State Pigeon hole Principle with an example.



- (31) Let f & g be functions from $\{1, 2, 3, 4\}$ to $\{a, b, c, d\}$ and from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ respectively with $f(1) = d$, $f(2) = c$, $f(3) = a$ and $f(4) = b$ and $g(a) = 2$, $g(b) = 1$, $g(c) = 3$ and $g(d) = 2$
- if f & g one to one?
 - if f and g on-to?
 - Does either f or g have an inverse? If so find this inverse.

- (32) Find the Solution of the recurrence relation
 $a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}$ if $a_0 = 2$, $a_1 = 2$, $a_2 = 4$

- (33) Define an Euler Circuit & Euler path in an Undirected graph. Compute the maximal flow from the following network flow.



- (34) Find the Solution to the System of Congruences
 $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{5}$, $x \equiv 3 \pmod{7}$ Using Chinese Remainder theorem

- (35) What are the necessary Conditions for a graph to be isomorphic? List & illustrate with an example.

- (36) Let R be a relation defined on set of natural numbers N , Such that $a, b \in N$, $aRb \iff ab = 2^k$, Where $k \in \{0, 1, 2, \dots\}$. Show that R is a partial ordering relation on N .

- (37) Write the type of fuzzy Set operations with their definition.

(5)

(38) Define Chromatic number. How does Kruskal's algorithm find minimum Spanning tree.

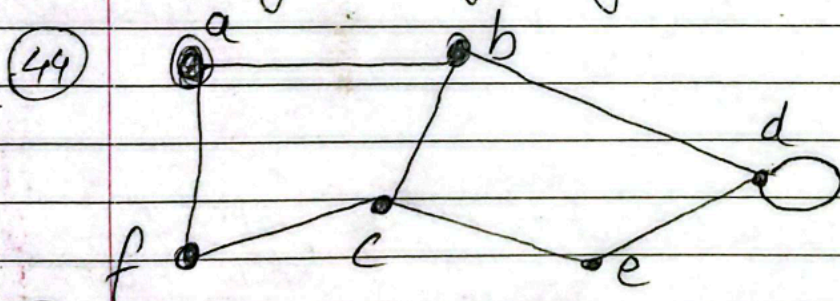
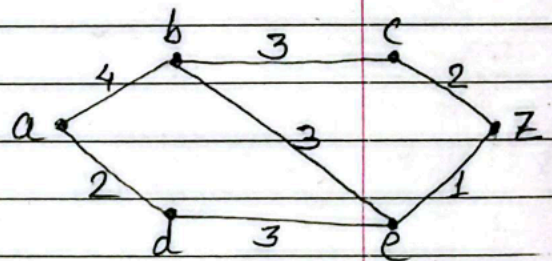
(39) When do we use permutation rather than combination? How many 5-digit numbers can be generated using the digits 0 to 9, if each number starts with 98 and no digit appears more than once.

(40) Explain fuzzy set with example. How do you find Complement of a fuzzy set?

(41) What is Congruent modulo? Determine whether 37 is Congruent to 3 modulo 7 and whether -29 is Congruent to 5 modulo 17.

(42) Define network flow with example. What are Saturated edge, unsaturated edge and Slack value?

(43) What is Shortest path problem? Use Dijkstra's Shortest path algorithm to find the Shortest path between A & Z in the given weighted graph.



Represent the following graph using Adjacency matrix.

(45) Define the types of Tree Traversal Techniques. With example. Hint: DFS (Inorder, preorder, post order) & BFS.

THE END